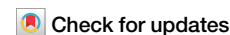




Classification of magnetism and altermagnetism in quasicrystals



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Altermagnetism (AM), an unconventional magnetic phase characterized by zero net magnetism protected by symmetry(s) other than parity-time ($\mathcal{PT}$) and a resulting spin-split band, has been studied exclusively in crystalline materials. Here, we extend the framework of AM to quasicrystals (QCs). We start from a comparison between the Néel state on the square lattice and that on a D_4 -symmetric Thue-Morse QC, with the latter obtained from solving a half-filled bipartite Hubbard model through the sign-problem-free projector quantum Monte Carlo algorithm. Consequently, although both Néel states belong to the same d -wave irreducible representation (IRRP) of the D_4 point group, they belong to different magnetic classes: The former Néel state is antiferromagnetism (AFM) protected by the combined $\mathcal{PT}$ and translational symmetry, while the lack of translational symmetry in the latter Néel state breaks the $\mathcal{PT}$ symmetry, and the additional mirror or rotation symmetry protects AM. This example suggests that AM is more common in QCs than in crystals and can be easily explored through a point-group symmetry-based classification. Therefore, we classify collinear magnetic phases in 2D D_n -symmetric QCs without spin-orbit coupling, by using IRRPs of D_n . Consequently, the identity IRRP represents ferromagnetism, the inversion-odd 1D IRRPs for twice-of-odd n represent AFM, and all the remaining 1D IRRPs represent AM, protected by either mirror or rotation symmetry. We further take the Hubbard model to verify this result in various QCs with different symmetries. Our work highlights the QC as a natural platform where AM is common among magnetic phases.

Altermagnetism (AM)^{1–3} has recently emerged as a distinct class of magnetic order^{4–17}, characterized by two key properties: zero net magnetization similar to antiferromagnetism (AFM), and a spin-split electronic band structure characteristic of ferromagnetism (FM). This duality makes AM a promising candidate for next-generation spintronic applications^{15,18–23}, enabling both efficient spin-current manipulation and robust magnetic control without the undesirable stray fields associated with FM.

Symmetry-based classification of magnetic states^{24–29} is crucial for the exploration of AM. From the aspect of symmetry, AM is a magnetic phase without combined parity ($\mathcal{P}$) and time-reversal ($\mathcal{T}$) symmetry, while exhibiting zero net magnetism protected by other symmetries. AM and other unconventional magnetic orders^{30,31} are systematically classified using the spin-space group, which incorporates both spatial and spin symmetries in the absence of spin-orbital coupling (SOC)^{24–28}. From a representation theory perspective, AM corresponds to a 1D, non-trivial, inversion-even irreducible representation (IRRP) of the magnetic order parameter^{1,2,32}, which breaks the combined $\mathcal{PT}$ symmetry.

The material realization of AM is the recent research focus. AM can arise from various mechanisms, including the presence of nonmagnetic atoms^{2,33}, spontaneously broken orbital order^{34–37}, or Pomeranchuk instabilities^{30,31,38,39}. Engineered routes to AM have been proposed in several platforms. For example, twisted van der Waals ferromagnets can host moiré superlattices with spin-split bands^{40–42}, and inversion-symmetry-broken systems such as ferroelectrics can also exhibit AM^{43–49}. Furthermore, external tuning through strain, lattice vacancies, or the formation of spin clusters has been shown to stabilize extrinsic AM^{50–54}. In the search for intrinsic AM materials, although many materials have been theoretically predicted to be AM, only a few of them are supported by experimental evidences^{55–60}, with the confirmed ones even rarer. While previous studies of AM are mostly focused on conventional crystalline materials, here we demonstrate that the quasicrystal (QC) can serve as a natural platform where intrinsic AM is more common and more easily explored.

The QC is a unique long-range-ordered structure without translational symmetry. In QCs, the aperiodic structure and the higher-fold rotational symmetries^{61–63} produce various novel phases^{64–96}, including the magnetic

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phases^{64,66,68,69,82,83,91–93}. Now an interesting question arises: Is AM a common magnetic phase in the QC? We shall show that the unique aperiodic structure of QC renders AM more common and more easily explored in QCs than in crystals. Therefore, it is now urgent to classify magnetic phases in QCs based on symmetry to guide the exploration of AM in QCs. Unlike crystals, the QCs lack translational symmetry, making the classification of magnetic phases quite different.

In this paper, we investigate AM driven by interaction in 2D QCs without SOC. We start by showing the difference between the Néel state on the square lattice and that on a D_4 -symmetric Thue-Morse QC. Here, the latter Néel state is obtained from solving a half-filled bipartite Hubbard model by the sign-problem-free projector quantum Monte Carlo (PQMC) algorithm^{97–99}. Consequently, both Néel states belong to the d_{xy} -wave IRRP of D_4 . However, while the former Néel state is AFM protected by the combined $\mathcal{PT}$ and translational symmetry, the lack of translational symmetry in the latter Néel state breaks the $\mathcal{PT}$ symmetry, and the additional mirror or rotation symmetry protects zero total magnetism, leading to AM. This example demonstrates that AM is more common in QCs than in crystals and is more easily explored through a point-group symmetry-based classification. Therefore, going beyond focusing on some specific examples of AM in QCs like that in refs. 91,92, we classify collinear magnetic phases in 2D D_n -symmetric QCs, based on 1D IRRPs of D_n , including A_1 , A_2 , B_1 , and B_2 . Symmetry analysis suggests that, while the identity IRRP A_1 represents FM and the inversion-odd $B_{1,2}$ IRRPs for twice-of-odd n represent AFM, all the remaining 1D IRRPs represent AM, protected by the mirror-reflection or the rotation symmetry. For verification, we take the Hubbard model as an example and the solved magnetic states in various QCs with different symmetries are all well described by the classification. Our work sets the QC as a natural platform for intrinsic AM, thereby providing a theoretical foundation for future experimental investigations and potential applications. We also discuss some properties of AM QC materials.

Results

AM in the QC

We start by a comparison between the Néel states on two D_4 symmetric lattices, i.e. the square lattice shown in Fig. 1(a) and the Thue-Morse QC shown in Fig. 1(b). The Néel state on the square lattice has been fully studied^{100–103}, with Fig. 1(a) exhibiting an exemplary result. To explore the Néel state on the Thue-Morse QC, we study the following half-filled bipartite Hubbard model by the sign-problem-free PQMC algorithm^{97–99},

$$H = -t_1 \sum_{\langle i,j \rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) - t_2 \sum_{\langle\langle i,j \rangle\rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (1)$$

Here, the t_1 and t_2 terms represent the hoppings along the nearest-neighbor and the next-nearest-neighbor bonds, respectively, as shown in Fig. 1(b). We set $t_1 = 1$, $t_2 = 0.5$ and $U = 8$ in our PQMC calculation. See ref. 104 for details of this method. The obtained magnetic pattern is shown in Fig. 1(b). Clearly, only symmetry related sites host equal amplitude of the magnetic moment. If we consider different U , we get similar magnetic pattern exhibiting Néel order, see Supplementary Information (SI)¹⁰⁵.

Obviously, the magnetic patterns on both lattices show similar d_{xy} -wave (i.e. the B_2 IRRP) symmetry behaviors under the D_4 point group operations around their symmetry centers marked by the solid pentagrams: both are unchanged under the inversion or mirror reflection about the diagonal directions and change signs under the c_4^1 rotation or mirror reflection about the x - or y -axes σ_{xy} . Although both magnetic patterns are inversion-even and hence $\mathcal{PT}$ -odd, the two Néel states belong to different magnetic classes: In Fig. 1(a), the $\mathcal{PT}$ symmetry is recovered after the translation along a unit vector, leading to AFM; while in Fig. 1(b), the lack of translational symmetry breaks the $\mathcal{PT}$ symmetry, and the combined $\mathcal{T}$ and c_4^1 or σ_{xy} symmetry which guarantees zero net magnetism identifies AM.

The fact that the two Néel states shown in Fig. 1(a) and (b) behave differently under $\mathcal{PT}$ can be understood from a different but equivalent perspective, clarified here. In Fig. 1(a), the square lattice actually hosts three other unequivalent inversion centers marked B, C, and D. One has the flexibility to set B or C as the inversion center, under which the Néel state is inversion-odd and hence $\mathcal{PT}$ -symmetric. However, in Fig. 1(b), the aperiodic structure forbids another inversion center: If a QC hosts two inversion centers, then successive inversion operations around the two centers will lead to a translational symmetry, which is forbidden. Therefore, Fig. 1(b) can host only one inversion center, under which the Néel state is inversion-even and hence breaks the $\mathcal{PT}$ symmetry. Note that in Fig. 1(b), only symmetry related sites host equal amplitude.

The AM in QCs can be characterized by the spin-resolved angle-resolved photoemission spectrum $\mathcal{A}_{\uparrow,\downarrow}(\mathbf{p}, \omega)$ ^{91,92,106}. $\mathcal{A}_{\uparrow,\downarrow}$ indicates the spectrum weight for spin $\uparrow$ or $\downarrow$ at a certain energy level ω for the detecting momentum $\mathbf{p}$, see *methods* for details. Experimentally, this can be measured by the spin-resolved angle-resolved photoemission spectroscopy experiments, which can be realized in QCs like in crystals. For a gapped state like the half-filled Néel state shown in Fig. 1(b), ω can be set as a certain energy below the Fermi level. In our study, the $\mathcal{A}_{\uparrow,\downarrow}(\mathbf{p}, \omega)$ is obtained through diagonalizing the magnetic-ordered mean-field Hamiltonian with the magnetic pattern input from the PQMC calculation, see *methods*. The obtained spectrum difference $\mathcal{A}_\uparrow - \mathcal{A}_\downarrow$ as a function of $\mathbf{p}$ for $\omega = -0.33t_1$ is shown in Fig. 1(c), which exhibit nonzero values with d_{xy} symmetry, characteristic of AM. The spectrum functions here and shown later are calculated with systems including the symmetry center. For a real sample, although it may not include the symmetry center, it can always include many approximate symmetry center due to the self-similarity structure of QCs. Then the detected spectrum function may not be significantly affected

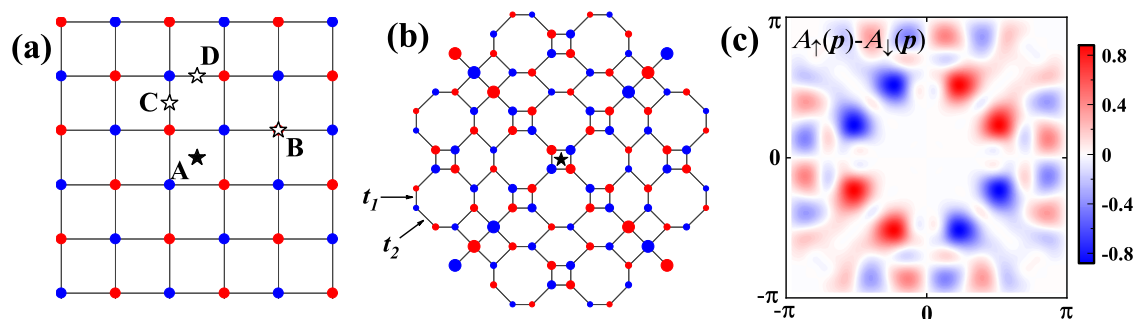


Fig. 1 | Comparison between Néel states on square lattice and Thue-Morse QC. **a** Néel state in square lattice. **b** Néel state in D_4 symmetric Thue-Morse QC. Here and in figures below, the red (blue) dots represent spin down (up), and the size of each dot represents amplitude of the magnetic moment. The solid pentagrams in (a, b)

indicate the D_4 center, and the hollow pentagrams in (a) indicate three additional inversion centers. **c** The spin-resolved spectral density difference $\mathcal{A}_\uparrow(\mathbf{p}, \omega) - \mathcal{A}_\downarrow(\mathbf{p}, \omega)$ for the AM state on the D_4 QC shown in (b), as a function of momentum $\mathbf{p}$ for $\omega = -0.33t_1$.

if the symmetry center is not included in the sample. For imperfect samples, the results may not be significantly affected either if the structure is approximately perfect.

This example suggests that AM is more common in QCs than in crystals, because QCs cannot provide additional translation operations to recover the broken $\mathcal{PT}$ symmetry, allowing for the possibility of AM protected by other point-group symmetries. Furthermore, since a QC can host at most one inversion center, it is convenient to explore AM through a point-group symmetry-based classification in the QC, which we shall perform below.

Magnetism classification in QCs

Magnetic orders in QCs without SOC are classified according to IRRPs of their point groups. The absence of SOC allows for the symmetry operation to act only on the coordinate degree of freedom, without altering the spin. This work specifically focuses on the 1D IRRPs, which describe collinear magnetic orders. Here, we only consider 2D QCs hosting the D_n point group. The magnetism classification according to 1D IRRPs of D_n is listed in Table 1.

For even n , the D_n group possesses four 1D IRRPs: A_1 , A_2 , B_1 and B_2 . A_1 is the identity IRRP whose basis function is invariant under all symmetry operations of D_n ; it represents for the s -wave magnetic phase. Both B_1 and B_2 IRRPs change sign under a c_n^1 rotation, and indicate magnetic phases with the highest angular momentum $l = n/2$. They are distinguished by their behavior under mirror reflection σ_x about the x -axis: B_1 is invariant while B_2 reverses sign. Examples of B_1 and B_2 IRRPs include the two different g - (h -) wave states in the D_8 (D_{10}) QC. A_2 is the product IRRP of B_1 and B_2 ; its basis function is invariant under c_n^1 but reverses sign under the mirror reflection. Example of A_2 includes the $g * g'$ - ($h * h'$ -) wave states in the D_8 (D_{10}) QC.

For odd n , the D_n group hosts just two 1D IRRPs: A_1 and A_2 . The corresponding basis functions exhibit similar behaviors under the symmetry operations to those in the case of even n . For example, the A_2 IRRP of D_5 includes the $h * h'$ -wave magnetic state.

The properties of the magnetic phases corresponding to each IRRP and their classification is clarified below.

For the identity IRRP A_1 phase, the spin remains invariant under all symmetry operations, resulting in no restriction on the total magnetism. This phase usually hosts a non-zero net magnetism and is FM.

For phases corresponding to non-identity 1D IRRPs, the spin flips under certain symmetry operations, ensuring zero net magnetism. These phases are classified according to their representations under the $\mathcal{PT}$ operation. Since a QC can host at most one inversion center, the $\mathcal{PT}$ -even (odd) is equivalent to inversion-odd (even) about this center, convenient for classification.

From the inversion parity $\mathcal{P}$ listed in Table 1, it is found that only the B_1 and B_2 IRRPs for twice-of-odd n are inversion-odd which conserve the $\mathcal{PT}$ symmetry and thus represent AFM, and all the remaining non-identity 1D IRRPs are inversion-even which break $\mathcal{PT}$ and represent AM. The AM phases include the A_2 IRRPs for all n and the $B_{1,2}$ IRRPs for twice-of-even n . In the former case, the AM is protected by mirror-reflection combined with $\mathcal{T}$; in the latter case, both the mirror-reflection and the c_n^1 rotation combined with $\mathcal{T}$ can protect AM.

The above results are summarized as: **In a 2D D_n -symmetric QC, the identity IRRP A_1 represents FM; the inversion-odd IRRPs $B_{1,2}$ for twice-of-odd n represent AFM; and all the remaining 1D IRRPs represent AM, protected by mirror reflection or rotation combined with $\mathcal{T}$ symmetries.**

In the following, we show various exemplifying states corresponding to the different situations listed in Table 1, which are all well described by the classification.

Examples

We take the following Hubbard model as an example to investigate magnetic phases in 2D QCs,

$$H = H_{\text{TB}} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (2)$$

$$H_{\text{TB}} = - \sum_{i \neq j, \sigma} e^{-r_{ij}/a} c_{i\sigma}^\dagger c_{j\sigma} - \mu \sum_{i, \sigma} n_{i\sigma}.$$

Here, H_{TB} is the TB Hamiltonian, r_{ij} is the distance between the sites i and j , the length unit a for each studied QC is given in *methods*, μ is the chemical potential, $n_{i\sigma}$ is the electron number operator for spin- σ and site i , and U represents the Hubbard repulsion. Open boundary condition is imposed here. We shall not adopt the PQMC approach to solve this model due to the presence of the sign problem. To solve this model, we adopt the following linear-response theory detailed in *methods*, which can effectively avoid the problem of being trapped at local energy minima, which is usually encountered in the mean-field study^{91,92}.

Considering collinear magnetism with fixed polarization orientation, the real-space distribution of the relative magnetic moment $\{\Delta_i\}$ is solved as the eigenvector of the following eigenvalue problem of the spin susceptibility χ for U approaching its critical value from below,

$$\sum_j \chi_{ij} \Delta_j = \lambda \Delta_i. \quad (3)$$

For a given set of Hamiltonian parameters, the realized $\{\Delta_i\}$ is that corresponds to the largest eigenvalue λ . As χ inherits the point group symmetry of the QC, the eigenvector(s) corresponding to the same eigenvalue form(s) an IRRP of the point group. By tuning the Hamiltonian parameters, we can

Table 1 | Classification of magnetic phases in 2D D_n -symmetric QCs without SOC according to the 1D IRRPs of D_n

n	IRRP	c_n^1	σ_x	$\mathcal{P}$	Basis functions and symmetry class	Zero magnetism	$\mathcal{PT}$	Magnetism
2*odd	A_1	1	1	1	1, s	×	− 1	FM
	A_2	1	− 1	1	$\text{Re}((x + iy)^{n/2}) * \text{Im}((x + iy)^{n/2})$, e.g. $h * h'$ for $n = 10$	✓	− 1	AM
	B_1	− 1	1	− 1	$\text{Re}((x + iy)^{n/2})$, $l = n/2$, e.g. $h_{x^5-10x^3y^2+5xy^4}$ for $n = 10$	✓	+ 1	AFM
	B_2	− 1	− 1	− 1	$\text{Im}((x + iy)^{n/2})$, $l = n/2$, e.g. $h'_{5x^4y-10x^2y^3+y^5}$ for $n = 10$	✓	+ 1	AFM
2*even	A_1	1	1	1	1, s	×	− 1	FM
	A_2	1	− 1	1	$\text{Re}((x + iy)^{n/2}) * \text{Im}((x + iy)^{n/2})$, e.g. $g * g'$ for $n = 8$	✓	− 1	AM
	B_1	− 1	1	1	$\text{Re}((x + iy)^{n/2})$, $l = n/2$, e.g. $g_{x^4-6x^2y^2+y^4}$ for $n = 8$	✓	− 1	AM
	B_2	− 1	− 1	1	$\text{Im}((x + iy)^{n/2})$, $l = n/2$, e.g. $g'_{4x^3y-4xy^3}$ for $n = 8$	✓	− 1	AM
odd	A_1	1	1	−	1, s	×	−	FM
	A_2	1	− 1	−	$\text{Re}((x + iy)^n) * \text{Im}((x + iy)^n)$, e.g. $h * h'$ for $n = 5$	✓	−	AM

D_n possesses the generating element c_n^1 (rotation) and mirror reflection (σ_x). The inversion parity $\mathcal{P}$ only applies in cases of even n . $l = n/2$ represents the angular momentum. The “✓” or “×” indicates that the property is satisfied or not.

Fig. 2 | Examples of FM states. **a** The magnetic order of the example in Thue-Morse QC. **b** The magnetic order of the example in Penrose QC.

(a) s -wave for D_4 **(b) s -wave for D_5**

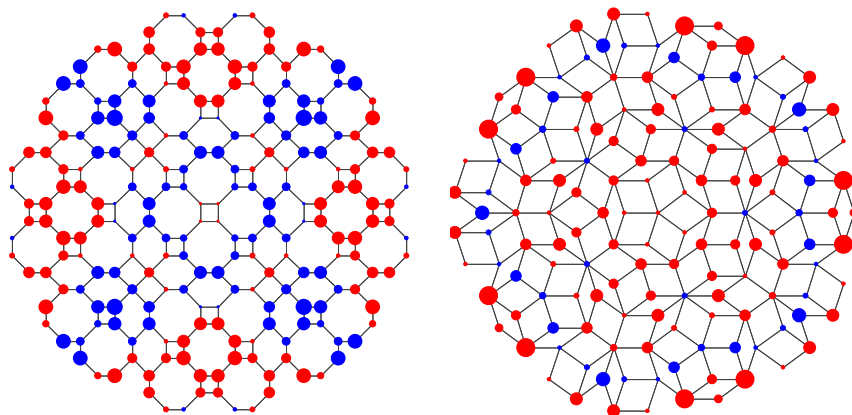
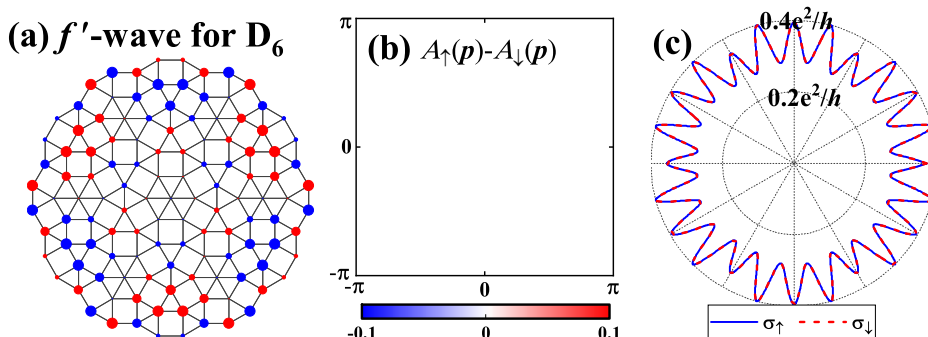


Fig. 3 | An example of AFM state in a D_6 QC. **a** The magnetic order of the example. **b** The spin-resolved spectral density difference $\mathcal{A}_\uparrow(\mathbf{p}, E_F) - \mathcal{A}_\downarrow(\mathbf{p}, E_F)$ as a function of momentum $\mathbf{p}$, showing zero result. **c** Spin conductance $\sigma_\uparrow$ and $\sigma_\downarrow$ as functions of the angle ϕ with $R = 3.5$.



realize different magnetic phases corresponding to all 1D IRRPs listed in Table 1.

As the obtained magnetic-ordered state is generally gapless here, to characterize AM, we consider the spin-resolved angle-resolved photoemission spectrum $\mathcal{A}_\uparrow(\mathbf{p}, \omega) - \mathcal{A}_\downarrow(\mathbf{p}, \omega)$ introduced previously, as well as the spin conductance $\sigma_{\uparrow,\downarrow}(\phi)$ ⁹¹. Here ω is set to the Fermi level. The spin conductance $\sigma_{\uparrow,\downarrow}$ can be measured by placing two tips perpendicular to the QC at $(R \cos \phi, R \sin \phi)$ and $(-R \cos \phi, -R \sin \phi)$, and the measured value is a function of the angle ϕ , plotted in the polar coordinate in the following. Experimentally, this can be detected by double-tip scanning tunneling microscopy experiments. This measurement can be realized in QCs like in crystals. This may be also robust in finite-size or imperfect samples like the spin-resolved spectrum difference. $\mathcal{A}_\uparrow - \mathcal{A}_\downarrow$ and $\sigma_{\uparrow,\downarrow}$ are obtained through the magnetic-ordered mean-field Hamiltonian with the magnetic pattern input from the linear-response theory, see *methods*.

To illustrate the classification, we provide examples for each case outlined in Table 1. Specifically, $n = 10, 6$ are chosen for twice-of-odd n , $n = 8, 4$ are chosen for twice-of-even n , and $n = 5$ is chosen for odd n . In each case, we tune Hamiltonian parameters (see SI¹⁰⁵ for details) to obtain magnetic states corresponding to all 1D IRRPs through solving Eq. (3). The magnetic patterns and the corresponding spin-resolved spectrum difference at the Fermi level $\mathcal{A}_\uparrow(\mathbf{p}, E_F) - \mathcal{A}_\downarrow(\mathbf{p}, E_F)$ and spin conductance $\sigma_\uparrow$ and $\sigma_\downarrow$ for all these cases are shown in SI¹⁰⁵. Consequently, all these examples are well described by the classification. Below, we show examples for the FM, AFM and AM, respectively.

For the FM, Table 1 suggests that the s -wave states corresponding to the identity IRRP A_1 for all n are FM. We show some typical s -wave states obtained for $n = 4, 5$ corresponding to even and odd n , respectively. As shown in Fig. 2, these states do not change under any symmetry operation in the point group, leading to no restriction to the total magnetism. Therefore,

the net magnetism of these states is generally non-zero, leading to FM, verifying Table 1.

For the AFM, Table 1 suggests that the highest-angular-momentum states corresponding to the $B_{1,2}$ IRRPs for twice-of-odd n are AFM. We show a typical f' -wave (i.e. B_2) state for $n = 6$. As shown in Fig. 3 (a), this state is $\mathcal{P}$ odd and thus $\mathcal{PT}$ even, which protects the zero net magnetism. The spin-resolved spectrum difference $\mathcal{A}_\uparrow(\mathbf{p}, E_F) - \mathcal{A}_\downarrow(\mathbf{p}, E_F)$ (see Fig. 3 (b)) and spin conductance difference $\sigma_\uparrow - \sigma_\downarrow$ (see Fig. 3 (c)) for this state are zero, indicating that this state is AFM, verifying Table 1.

For the AM, Table 1 suggests that the states corresponding to IRRP A_2 for all n and IRRPs $B_{1,2}$ for twice-of-even n are AM. For the former case, we show a typical $h * h'$ -wave state for $n = 10$, a typical $d * d'$ -wave state for $n = 4$, and a typical $h * h'$ -wave state for $n = 5$. For the latter case, we show a typical d' -wave state for $n = 4$. As shown in Fig. 4 (a-d), these states are $\mathcal{P}$ even or do not have $\mathcal{P}$ symmetry, breaking the $\mathcal{PT}$ symmetry. However, all these states have zero net magnetism protected by mirror reflection. Additionally, the zero net magnetism in the latter case is also protected by c_4^1 rotation. The spin resolved spectrum difference $\mathcal{A}_\uparrow(\mathbf{p}, E_F) - \mathcal{A}_\downarrow(\mathbf{p}, E_F)$ and spin conductance difference $\sigma_\uparrow - \sigma_\downarrow$ are non-zero and have the same symmetry as the corresponding state: Fig. 4 (e-f) show the results for the $h * h'$ -wave state for $n = 10$ as an example, and those for the remaining states are shown in SI¹⁰⁵, which are similar, indicating that these states are AM, verifying Table 1.

Some properties of AM QC materials

Here, we discuss some properties of AM QC materials. One is the proximity effect¹⁰⁷. In crystals, it is found that in a heterostructure including a paramagnetic material and an AM material, the real-space symmetry of the magnetic order in the AM material can be imprinted onto the paramagnetic material¹⁰⁷. In a heterostructure including a paramagnetic material and a QC exhibiting AM, a real-space magnetic pattern with the symmetry of the

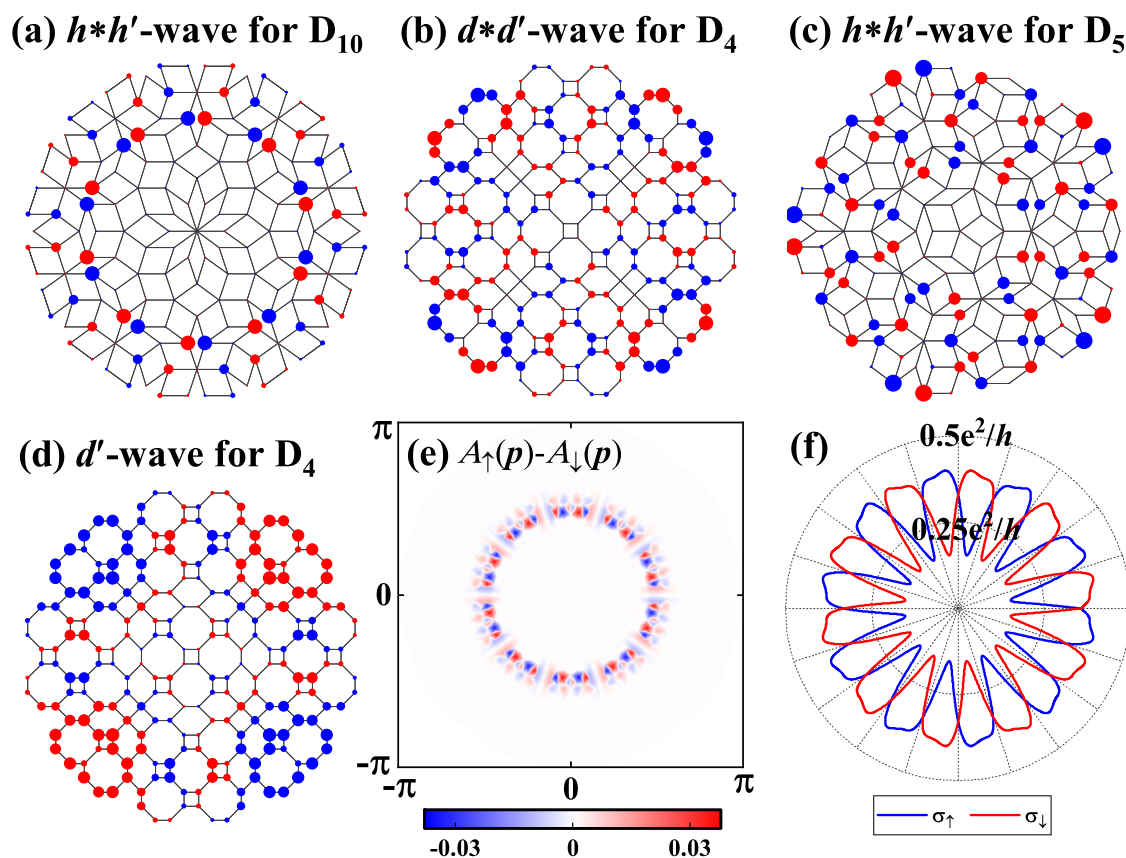


Fig. 4 | Examples of AM states. **a** The magnetic order of the example in decagonal QC. **b** The magnetic order of an example in Thue-Morse QC. **c** The magnetic order of the example in Penrose QC. **d** The magnetic order of another example in Thue-

Morse QC. **e** The $A_{\uparrow}(p, E_F) - A_{\downarrow}(p, E_F)$ as a function of momentum p for the AM state in the D_{10} QC shown in (a). **f** The $\sigma_{\uparrow}$ and $\sigma_{\downarrow}$ as functions of the angle ϕ with $R = 4.5$ (f), for the AM state in the D_{10} QC shown in (a).

IRRP of the point group of the one in the QC may be imprinted onto the paramagnetic material, which is similar to the proximity effect found in crystals¹⁰⁷, as illustrated in Fig. 5. This proximity effect may not be influenced by whether the system has translational symmetry.

The other is electrical control of electronic and spin transport properties^{44,45,108,109}. In crystals, electrical control can be realized in systems in which AM coexists with ferroelectricity or antiferroelectricity by tuning the symmetry of the system^{44,45,108}. In QCs, since AM can exist and magnetic class can be changed by tuning the symmetry of the system, e.g. changing between an AFM state corresponding to IRRP $B_{1,2}$ of D_{2n} group and an AM state corresponding to IRRP A_2 of D_n group for odd n by tuning the symmetry of the lattice, electrical control may also be realized in systems in which AM coexists with ferroelectricity or antiferroelectricity like that in crystals^{44,45,108}. Since AM is more common in QCs than in crystals, it may be easier to find AM materials which can be electrical controlled in QCs.

Discussion

We have shown that AM is more common in QCs than in crystals due to the presence of point group symmetry and lack of translational symmetry, and have further classified magnetic phases in 2D D_n -symmetric QCs according to the 1D IRRPs of D_n . Our results suggest that all 1D non-identity IRRPs are AM, except the inversion-odd $B_{1,2}$ IRRPs for twice-of-odd n . This result is widely verified in the exemplifying study of the Hubbard model in various QCs with different symmetries.

This work primarily focuses on the collinear magnetic states governed by non-trivial 1D IRRPs of the D_n group in 2D QCs. These findings can be naturally extended to 3D QC systems, e.g. with I_h point group. Additionally, other collinear magnetic states transforming under higher-dimensional IRRPs may also be AM. Furthermore, higher-dimensional IRRPs in 2- or 3-spatial dimensions can lead to unconventional magnetic phases such as

nematic or chiral phases. A full investigation of these states will be the subject of future research.

We also discuss the possible proximity effect and the possible electrical control of electronic and spin transport properties in AM QC materials. These can be explored in future studies.

Methods

The tight-binding Hamiltonian

In our study of the Hubbard model on various exemplifying QCs through the linear-response theory, the TB Hamiltonian in Eq.(2) is given by:

$$H_{\text{TB}} = - \sum_{i \neq j, \sigma} e^{-r_{ij}/a} c_{i\sigma}^{\dagger} c_{j\sigma} - \mu \sum_{i, \sigma} n_{i\sigma}. \quad (4)$$

Here $c_{i\sigma}^{(\dagger)}$ annihilates (creates) a spin- σ electron at site i , r_{ij} is the distance between sites i and j , a is the length unit marked in Fig. 6, μ is the chemical potential, $n_{i\sigma}$ is the number operator for spin- σ electrons at site i .

The five QC structures we studied are shown in Fig. 6, including the Penrose QC (D_5 point group, shown in (a) panel), decagonal QC (D_{10} point group, shown in (b) panel), 45°-twisted bilayer square lattice (D_8 point group, shown in (c) panel), square-triangle tiling QC (D_6 point group, shown in (d) panel), and Thue-Morse QC (D_4 point group, shown in (e) panel). The length unit a for each QC is marked on the corresponding panel. Specifically, for the 45°-twisted bilayer square lattice, there is also an inter-layer distance d shown in Fig. 6 (c), which is set to $d = 2a$ in our calculations.

Linear response theory for the magnetic order

The magnetic order can be effectively obtained using linear response theory. We begin with a Hamiltonian perturbed by a small external magnetic field $\mathbf{h}_i$

along the z -direction,

$$H_h = H_0 + \sum_i \mathbf{h}_i \cdot \mathbf{S}_i. \quad (5)$$

Here, H_0 is the unperturbed Hamiltonian, $\mathbf{S}_i$ is the spin at site i , and $\mathbf{h}_i \rightarrow 0$ represents the external magnetic field. According to perturbation theory, the perturbed ground state $|\psi_G\rangle$ of H_h up to the first-order is expressed as

$$|\psi_G\rangle = |\psi_G^{(0)}\rangle + \sum_{n \neq G} \frac{\langle \psi_n^{(0)} | H_h | \psi_G^{(0)} \rangle}{E_G^{(0)} - E_n^{(0)}} |\psi_n^{(0)}\rangle, \quad (6)$$

where $|\psi_G^{(0)}\rangle$ and $E_G^{(0)}$ are the unperturbed ground state and energy of H_0 , and $|\psi_n^{(0)}\rangle$ and $E_n^{(0)}$ are the n -th excited eigenstates and eigenvalues of H_0 . To determine the magnetic order, we evaluate the ground state expectation

value of the spin operator at site i ,

$$\langle S_i \rangle = \langle \psi_G | S_i | \psi_G \rangle = \langle \psi_G^{(0)} | S_i | \psi_G^{(0)} \rangle + \sum_{j, n \neq G} \left(\frac{\langle \psi_G^{(0)} | S_i | \psi_n^{(0)} \rangle \langle \psi_n^{(0)} | H_h \cdot \mathbf{S}_j | \psi_G^{(0)} \rangle}{E_G^{(0)} - E_n^{(0)}} + \text{c.c.} \right) + O(h^2). \quad (7)$$

Here, $O(h^2)$ is higher-order contribution and can be ignored in the limit $\mathbf{h}_i \rightarrow 0$. The first term, $\langle \psi_G^{(0)} | S_i | \psi_G^{(0)} \rangle = 0$, vanishes since the unperturbed ground state of H_0 is spin SU(2) symmetric, while the spin operator $\mathbf{S}_i$ breaks this symmetry. Therefore, the leading order contribution is given by

$$\langle S_i \rangle = \sum_{j, n \neq G} \left(\frac{\langle \psi_G^{(0)} | S_i | \psi_n^{(0)} \rangle \langle \psi_n^{(0)} | H_h \cdot \mathbf{S}_j | \psi_G^{(0)} \rangle}{E_G^{(0)} - E_n^{(0)}} + \text{c.c.} \right). \quad (8)$$

By applying linear response theory, we can relate the induced spin polarization to the external field, $\langle S_i \rangle \propto \mathbf{h}_i$. Specifically, for a magnetic field $\mathbf{h}_i$ pointing along the z -direction, the induced spin polarization along z -axis, S_i^z , is proportional to the field strength, h_i^z . This relationship leads to the following expression,

$$\begin{aligned} \langle S_i^z \rangle &= \sum_{j, n \neq G} \left(\frac{\langle \psi_G^{(0)} | S_i^z | \psi_n^{(0)} \rangle \langle \psi_n^{(0)} | H_h^z S_j^z | \psi_G^{(0)} \rangle}{E_G^{(0)} - E_n^{(0)}} + \text{c.c.} \right) \\ &= \sum_j 2\alpha * \text{Re} \left(\sum_{n \neq G} \frac{\langle \psi_G^{(0)} | S_i^z | \psi_n^{(0)} \rangle \langle \psi_n^{(0)} | S_j^z | \psi_G^{(0)} \rangle}{E_G^{(0)} - E_n^{(0)}} \right) \langle S_j^z \rangle. \end{aligned} \quad (9)$$

From Eq. (9), the magnetic order, represented by the vector of spin expectation values, corresponds to an eigenvector of the matrix, defined by the term within the parenthesis.

To analyze the magnetic order, we first introduce the Matsubara Green's function defined by,

$$\mathcal{G}_{ij}(\tau) = \langle T_\tau S_i^z(\tau) S_j^z(0) \rangle, \quad (10)$$

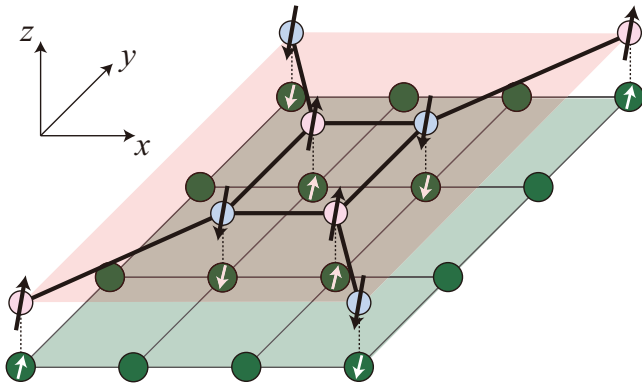


Fig. 5 | Illustration of possible proximity effect. The light pink layer represents an AM QC layer and the green layer represents a paramagnetic layer.

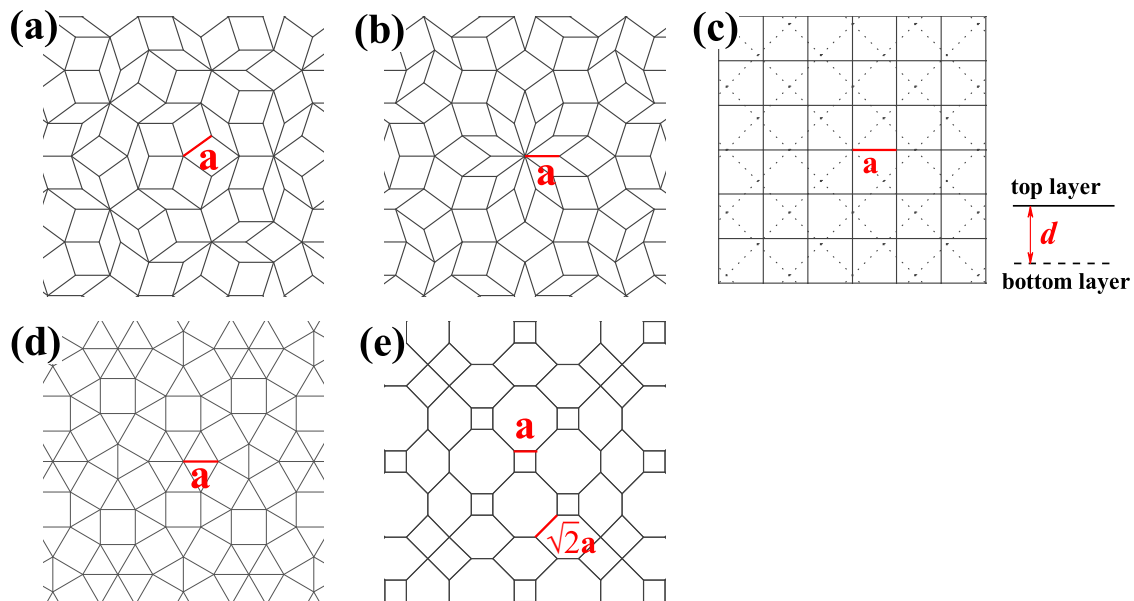


Fig. 6 | The five QC structures we studied, where the red line marks the length unit $a = 1$. **a** Penrose QC with D_5 symmetry. **b** decagonal QC with D_{10} symmetry. **c** 45°-twisted bilayer square lattice with D_8 symmetry. The solid(dashed) line represents

the top(bottom) layer. The interlayer distance d is marked at the bottom right corner. **d** square-triangle tiling QC with D_6 symmetry. **e** Thue-Morse QC with D_4 symmetry.

where τ is the imaginary time and T_τ represents the time-ordered product of operators. Its Fourier transform gives the static spin susceptibility χ_{ij} ,

$$\chi_{ij} \equiv \mathcal{G}_{ij}(i\omega_n = 0), \quad (11)$$

$$\mathcal{G}_{ij}(i\omega_n) = \int_0^\beta d\tau \mathcal{G}_{ij}(\tau) e^{i\omega_n \tau},$$

where β is the inverse temperature and $\omega_n = \frac{2\pi n}{\beta}$ is the Matsubara frequency. The Lehmann's representation of χ_{ij} is precisely the summation in the bracket in Eq. (9). Therefore, if χ_{ij} is seen as a matrix element where i and j are the row and column indices, respectively, the magnetic order is obtained from the eigenvector of the real part of χ matrix.

The spin susceptibility can be obtained as follows. First we obtain the bare susceptibility $\chi^{(0)}$, whose definition is similar to the above but the expectation is obtained over the bare TB Hamiltonian. Then the expression of $\chi^{(0)}$ is

$$\chi_{ij}^{(0)} = \sum_{mn} \xi_{im} \xi_{in} \xi_{jm} \xi_{jn} \frac{n_F(\epsilon_m - \mu) - n_F(\epsilon_n - \mu)}{\epsilon_n - \epsilon_m}, \quad (12)$$

where n_F represents the Fermi-Dirac distribution function, ϵ_m is the m -th eigenvalue of H_{TB} , ξ_{im} is the i -th value in the eigenvector of H_{TB} belonging to ϵ_m , and μ is the chemical potential. Then the renormalized spin susceptibility for the interacting case up to the random-phase approximation level is obtained from the Dyson's equation,

$$\chi = (I - U\chi^{(0)})^{-1} \chi^{(0)}. \quad (13)$$

When the interaction strength is larger than the critical value U_c , χ diverges, forming a magnetic order. The magnetic pattern is obtained from the eigenvector belonging to the largest eigenvalue of χ for $U \rightarrow U_c^-$. For this single-orbital case, U in the Dyson's equation is just a number and equals the strength of the onsite intraorbital Coulomb repulsion. Thus, χ commutes with $\chi^{(0)}$.

Characterization of altermagnetism

The AM in QCs can be characterized by the spin-resolved angle-resolved photoemission spectrum^{91,92,106} and the spin conductance⁹¹.

To obtain these quantities, we start with a mean-field Hamiltonian

$$H_{MF} = H_{TB} + \sum_i \Delta_i (n_{i\uparrow} - n_{i\downarrow}). \quad (14)$$

In our study on the Hubbard model on the D_4 -symmetric Thue-Morse QC through the PQMC approach, we adopt $\Delta_i = -\frac{2U}{3} S_i^z$, with $\{S_i^z\}$ given by the PQMC result. In our study on the Hubbard model on various QCs through the linear-response theory, we adopt $\{\Delta_i\}$ as the magnetic pattern obtained by solving Eq. (3) in the main text. We note that, when showing the results of the linear-response theory based study, only the region near the rotational center are shown in all figures in the main text and here.

In the Wannier basis, the matrix form of H_{MF} is a block diagonal matrix,

$$(H_{MF}) = \begin{pmatrix} (H_\uparrow) & 0 \\ 0 & (H_\downarrow) \end{pmatrix}. \quad (15)$$

Here, (H_{MF}) is the matrix form of H_{MF} in the Wannier basis, and (H_σ) is the matrix form of the mean-field Hamiltonian for spin σ ($\sigma = \uparrow, \downarrow$). Then we solve the eigenvalues and eigenvectors of (H_σ)

$$\sum_j (H_\sigma)_{ij} \xi_{jm}^\sigma = E_m^\sigma \xi_{im}^\sigma, \quad (16)$$

where E_m^σ and ξ_{im}^σ are the m -th eigenvalue and eigenvector of (H_σ) , respectively.

We consider the spectrum weight for spin σ at a certain energy level ω , which is expressed as

$$\mathcal{A}_\sigma(\mathbf{p}) = -\frac{\eta}{\pi} \frac{|\langle \mathbf{p}\sigma | m\sigma \rangle|^2}{(\omega - E_m^\sigma)^2 + \eta^2}, \quad (17)$$

where $|m\sigma\rangle$ is the m -th eigenstate of H_σ , $|\mathbf{p}\sigma\rangle = \frac{1}{\sqrt{N}} \sum_i e^{i\mathbf{p}\cdot\mathbf{r}_i} c_{i\sigma}^\dagger |0\rangle$ (N is site number, $\mathbf{r}_i$ is the position of site i and $|0\rangle$ is vacuum state) is the plane wave with momentum $\mathbf{p}$, and η is a small Lorentzian expansion width. For Fig. 1 (c), we set $\omega = -0.33t_1$ relative to the Fermi energy E_F . For other calculations, we set $\omega = 0$ relative to E_F .

The spin conductance is calculated by the Landauer-Büttiker formula¹¹⁰⁻¹¹² and the recursive Green's function method^{113,114}, see ref. 91.

Data availability

All data are displayed in the main text and the supplementary information.

Code availability

The code that supports the plots within this paper is available from the corresponding author upon request.

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Author contributions

F. Yang proposed the main idea and supervised the study. Z.-Y. Shao and Y.-B. Liu performed the linear-response theory based calculation. J.-H. Ji performed the quantum Monte-Carlo calculation. F. Yang, Z.-Y. Shao, J.-H. Ji, C. Lu, Z. Pan and Y.-B. Liu significantly contributed to the data analysis, discussion and paper writing.

Competing interests

The authors declare no competing interests.

Additional information

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